PM2.5 Forecasting Using Machine Learning

Project Report | Prototype and Validation Plan

Project type	Time-series regression / air-quality forecasting
Development dataset	Beijing Multi-Site Air Quality Data (UCI Repository)
Model explored	Random Forest Regressor
Target	PM2.5 concentration
Validation status	Prototype; data integrity and final metrics still require verification

Draft report. Insert final performance values only after the notebook has been rerun and data-quality checks have passed.

1. Abstract

This project develops a prototype workflow for estimating PM2.5 concentrations from historical air-quality and meteorological measurements. It combines data preparation, time-based feature engineering, a chronological train-test split, Random Forest regression, and evaluation against a previous-hour persistence baseline. The development dataset is the Beijing Multi-Site Air Quality dataset. The current work is a prototype and does not yet demonstrate reliable performance in European countries.

2. Problem Statement

PM2.5 consists of fine airborne particles relevant to air-quality monitoring and public health. Historical pollutant and weather observations can be used to investigate whether machine-learning methods can predict near-term PM2.5 concentrations. The goal is to build a reproducible forecasting pipeline and evaluate it on observations later in time than those used for training.

3. Dataset

The development dataset is the Beijing Multi-Site Air Quality Data hosted by the UCI Machine Learning Repository: https://archive.ics.uci.edu/dataset/501/beijing+multi+site+air+quality+data. It contains hourly observations from multiple monitoring stations and includes PM2.5, other pollutants, and weather variables.

Data-integrity note: during notebook development, the working dataframe was reported as 905,900 rows and 28 columns. This is larger than the commonly documented original dataset size, and the dataframe included columns such as predictions and evaluation results. The final analysis must rebuild a clean dataframe from source files, verify row counts and station/timestamp uniqueness, and exclude previously generated predictions or metrics from model inputs.

A separate 12-row file with columns START_TIME and VALUE was also examined. The pollutant represented by VALUE was not confirmed, so it is not treated as a valid PM2.5 dataset in this report.

4. Features and Target

Candidate predictors used in the prototype are temperature (TEMP), pressure (PRES), dew point (DEWP), rainfall (RAIN), wind speed (WSPM), SO2, NO2, CO, O3, PM10, hour, month, and PM2.5 from the previous hour. The target is PM2.5.

The previous-hour feature must be created separately for each station and must only refer to the immediately preceding hour. Time gaps and duplicate station/timestamp records must be handled explicitly.

5. Methodology

- Load the station CSV files and assign a consistent station identifier.
- Parse timestamps, sort by station and time, and inspect missing values and duplicate station/timestamp pairs.
- Create time-based predictors and a validated one-hour PM2.5 lag without a many-to-many merge.
- Split observations chronologically within each station: earlier 80% for training and later 20% for testing.
- Fit imputers and the Random Forest model using training data only.
- Generate predictions for the held-out test set.
- Evaluate MAE, RMSE, and R^2, and compare with a baseline that predicts the previous-hour PM2.5 value.
- Export verified predictions, metrics, plots, and the trained model.

6. Model and Evaluation

The model explored is scikit-learn's RandomForestRegressor. Mean Absolute Error (MAE) summarizes average absolute prediction error. Root Mean Squared Error (RMSE) gives larger errors more weight. R^2 measures model fit relative to a mean-target reference.

Metric	Final value	Interpretation
MAE	To be filled from verified run	Lower is better
RMSE	To be filled from verified run	Lower is better
R^2	To be filled from verified run	Closer to 1 is generally better
Baseline MAE	To be filled from verified run	Lower is better
Baseline RMSE	To be filled from verified run	Lower is better

No numerical performance results are asserted in this draft because the notebook's final outputs have not been verified. Populate this table from a clean, reproducible run only.

7. Limitations

- The current model is developed using Beijing data; transfer to European cities has not been demonstrated.
- The reported working dataframe size requires investigation against the original files.
- Duplicate station/timestamp rows can create misleading results and must be resolved.
- Using future-derived columns, prediction columns, or metrics as predictors would cause data leakage and must be prevented.
- Performance should be compared with a simple baseline on the same held-out observations.

8. Next Steps

- Restart the notebook kernel and run all cells from the original source files in order.
- Verify source row counts, station identifiers, timestamp uniqueness, missingness, and date range.
- Rebuild the lag feature safely and ensure it is truly one hour earlier.
- Use a chronological split and fit preprocessing only on the training partition.
- Record final metrics and save prediction plots and CSV outputs.
- Obtain documented European PM2.5 observations, align units and timestamps, and evaluate separately before claiming European forecasting capability.

9. References

UCI Machine Learning Repository. Beijing Multi-Site Air Quality Data. https://archive.ics.uci.edu/dataset/501/beijing+multi+site+air+quality+data

European Environment Agency. Datahub. https://www.eea.europa.eu/en/datahub

scikit-learn documentation. https://scikit-learn.org/

10. Conclusion

The notebook has been used to assemble a prototype PM2.5 forecasting workflow, including feature engineering, chronological splitting, Random Forest regression, error metrics, baseline comparison, plots, and export steps. The project should be considered a prototype until data integrity issues are resolved and final results are reproduced. European applicability requires separate validation using suitable European observations.

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